

Scalability & Future Plan

Date	07 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how OptiCrop can be scaled to support broader agricultural use, larger request volumes, additional crop and regional data, and more advanced decision-support capabilities. It also captures the team's roadmap for improving the machine-learning model, data sources, deployment architecture, and user experience.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Dependence on Manual Input Values	Recommendation quality depends on the accuracy of entered soil and climate values	High
2	Dataset and Model Scope	Predictions are limited to the crops and patterns represented in the current training dataset	High
3	No Real-Time Sensor or Weather Integration	Users must currently enter environmental parameters manually	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for current project-scale usage	Upgrade Render resources and use scalable production infrastructure as traffic grows
2	Data & Model	Uses the current crop recommendation dataset and trained model	Expand regional datasets, retrain models, and maintain model versioning
3	Performance	Prediction is performed by the Flask application	Optimize model loading, request handling, caching, and deployment resources
4	Data Integration	Seven parameters are entered manually	Integrate soil sensors, weather services, and automated data collection in future versions

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User Accounts & Prediction History	3 months	Allow users to securely save and review previous crop recommendations
Phase 3	Real-Time Weather & Soil Data Integration	6 months	Reduce manual entry and improve the relevance of input data
Phase 4	Regional Models, More Crops & Multilingual Support	9 months	Improve regional usefulness, crop coverage, and accessibility